

Yeritmary Rodríguez Delgado

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Profile

Robotics engineer and scientific researcher working across laboratory automation, machine learning, computer vision, and experimental imaging. My biological background and computational training allow me to follow a project from the scientific question through technical implementation, evaluate whether an engineering solution is biologically appropriate, and pivot when the technical approach no longer serves the experimental goal.

Education

Carnegie Mellon University

M.S. in Automated Science: Biological Experimentation

Pittsburgh, PA

Expected May 2027

- Relevant coursework: Machine Learning for Scientists, Automation of Scientific Research, Computational Genomics, Laboratory Methods for Automated Biology I & II.

University of Puerto Rico at Cayey

B.S. in Biology

Cayey, PR

June 2025

Professional Experience

Magnify Biosciences

Robotics Engineer

Pittsburgh, PA

2026–Present

- Develop and troubleshoot Python-based robotic liquid-handling workflows on the Opentrons Flex for automated experimental protocols.
- Integrate robotic hardware, labware geometry, liquid-handling parameters, and software control to improve workflow reliability and reproducibility.
- Diagnose hardware–software failures involving pipette positioning, aspiration and dispensing behavior, liquid sensing, tip handling, deck configuration, and labware calibration.
- Develop reusable automation utilities for dynamic tip selection, deck configuration, labware loading, and protocol execution.
- Develop an AI agent for laboratory automation to support automated error tracing, troubleshooting, and intelligent workflow assistance.

Carnegie Mellon University

Post-baccalaureate Research Fellow, Quantitative Imaging of Connexins

Pittsburgh, PA

Jul 2024–Jun 2025

- Applied image segmentation, fluorescence-intensity quantification, localization analysis, feature extraction, and morphological measurements across confocal microscopy datasets spanning 21 connexin isoforms.
- Quantified cell count, transfection efficiency, gap-junction plaque localization, and plaque morphology using Imaris and Cellpose workflows.
- Designed standardized image-analysis SOPs and reproducible workflows to improve consistency, documentation, and rigor across imaging experiments.
- Presented quantitative imaging research at ASBMB 2025 and PittCell 2025.

University of Puerto Rico at Cayey

NIH U-RISE Undergraduate Research Fellow

Cayey, PR

Aug 2021–Jul 2024

- Developed EthoVision-based behavioral analysis workflows to quantify movement and experimental outcomes across approximately

200 samples.

- Optimized data-acquisition parameters, experimental procedures, and analysis workflows to improve reproducibility of tracked behavioral measurements.
- Conducted behavioral experiments investigating shell selection, exploration, mobility, and responses to acute and chronic noise exposure in *Coenobita clypeatus*.

Selected Technical Projects

Automated Colorimetric Assay Optimization

2026

Closed-loop experimental automation combining OT-2 liquid handling, camera-based feedback, and active learning to iteratively optimize colorimetric formulations.

- Designed the experimental workflow and project documentation for a 96-well robotic color-mixing platform.
- Integrated robotic experimentation with vision-based feedback and machine-learning-driven experiment selection.

Uncertainty-Guided Nuclei Segmentation

2026

Active learning for scientific image segmentation under limited labeling budgets using U-Net models and Monte Carlo dropout.

- Compared passive sampling with uncertainty-guided acquisition on BBBC006 and BBBC039 microscopy datasets.
- Implemented U-Net segmentation, BCE + Dice loss, MC-dropout uncertainty estimation, and Dice/IoU evaluation.
- Refactored experiment code into a reusable Python package and documented reproducibility limitations and saved results.

Predicting Ixazomib Response in Cancer Cell Lines

2026

Biomedical machine-learning project integrating gene-expression and multimodal molecular data to investigate drug response in cancer cell lines.

- Applied Elastic Net regression, custom SVM/SVR implementations, biological feature engineering, PCA, and multimodal data integration.
- Evaluated Ixazomib-specific regression and sensitivity/resistance classification tasks.
- Collaborated with Jeffrey Yang, Lekhya Dommalapati, and Po-Wei Chang.

LabOS — Generative AI for Multi-Robot Laboratory Scheduling

2026

Generative-AI framework for scheduling and dynamic replanning across multi-robot scientific workflows.

- Designing scheduling and replanning strategies for heterogeneous laboratory robotic resources and experimental constraints.
- Exploring language-model-assisted planning for automated scientific workflows.

Computer Vision for Pipette Accuracy

2025

YOLOv8-based computer-vision pipeline for automated quality control of robotic liquid-handling performance.

- Developed object-detection workflows to assess dispensed liquid volume and identify inaccurate pipetting events from experimental imagery.
- Integrated computer vision with laboratory automation concepts for automated system validation.

Research Projects

Connexin Imaging and Gap-Junction Quantification

- Investigated connexin localization, plaque formation, morphology, and fluorescence behavior using confocal microscopy and quantitative image analysis.
- Worked with HEK293FT systems expressing multiple connexin isoforms, including Cx43 and Cx45.
- Applied segmentation, colocalization analysis, intensity quantification, and morphology-based feature extraction.

Hermit Crab Behavioral Research

- Conducted multi-year behavioral research investigating environmental noise effects on shell choice, exploration, mobility, and habituation.

- Used EthoVision video tracking and quantitative behavioral analysis to evaluate experimental outcomes.

Publications & Research Outputs

- GenBank accession MT889377, 2021.
- Manuscript in preparation: behavioral effects of environmental noise exposure in *Coenobita clypeatus*.

Presentations & Conferences

- Poster presentation, American Society for Biochemistry and Molecular Biology (ASBMB), 2025.
- Research presentation, PittCell, 2025.

Teaching & Leadership

Carnegie Mellon University

Pre-College Computational Biology Teaching Assistant

Pittsburgh, PA
2026

- Supported instruction in computational biology, scientific programming, and quantitative problem solving for pre-college students.

University of Puerto Rico at Cayey

President, Mathematics Circle

Cayey, PR

- Led student programming and mathematics-focused academic activities.

Technical Skills

Programming & Data	Python, SQL, PyTorch, NumPy, pandas, scikit-learn, matplotlib, Jupyter, Git, VS Code
Computer Vision	OpenCV, scikit-image, YOLOv8, U-Net, image segmentation, object detection, feature extraction, quantitative image analysis, ImageJ/Fiji, CellProfiler
Automation & Robotics	Opentrons Flex, Opentrons OT-2, Hamilton liquid-handling systems, Python Opentrons API, robotic liquid handling, hardware–software integration, Thermo Momentum
Machine Learning & AI	Deep learning, active learning, Elastic Net, SVM/SVR, uncertainty quantification, PCA, AI-agent development, model-guided experimental workflows
Imaging	Fluorescence microscopy, confocal microscopy, image acquisition, image-quality evaluation, intensity quantification, localization analysis, morphology analysis

Research Interests

Autonomous laboratories, scientific AI, robotic experimentation, computer vision, multimodal scientific models, experimental imaging, and intelligent laboratory systems.